

MATH 284 Final Review

- 1) Let $A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 4 & 6 & 8 \\ 3 & 2 & 1 & 0 \end{pmatrix}$.
 - a) Put A in reduced echelon form by hand.
 - b) If A is the augmented matrix for a linear system, is the system consistent? **Explain.**
 - c) Find all solutions to $A\mathbf{x} = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}$. Give your answer in vector form.
- 2) Let A be the matrix from (1). Find the following:
 - a) Rank A
 - b) A basis for $\text{Nul } A$
 - c) An orthogonal basis for $\text{Col } A$
 - d) The coordinates of $\begin{pmatrix} 1 \\ 2 \\ -1 \end{pmatrix}$ relative to the basis found in (c)
- 3) Find each of the following. Show all steps.
 - a) The area of a parallelogram with vertices at $(1, 2)$, $(3, 5)$, $(6, 3)$, and $(8, 6)$
 - b) The matrix of the transformation of $\mathbb{R}^2$ which first rotates clockwise by $\frac{\pi}{6}$ radians, and then expands horizontally by a factor of 4.
 - c) The image under the transformation from (b) of the parallelogram from (a)
 - d) The exact area of the transformed image from (c)
- 4) People who buy a domestic car (D) are 60% likely to buy a domestic car the next time, too. The other 40% will buy an imported car (I) the next time. People who buy an imported car are 75% likely to buy an imported car the next time, and 25% likely to buy a domestic car.

4) a) If 1000 customers each initially own a domestic car, how many will buy domestic for their 3rd car?

b) Find the **steady state vector**, and interpret it in the context of this problem.

5) Continuing (4), again assume that all 1000 people initially own domestic cars. Find an explicit formula for the exact number who will buy each type for their n^{th} car.

6) Evaluate $\begin{vmatrix} 1 & 2 & 3 \\ 0 & 1 & 2 \\ 2 & 1 & 2 \end{vmatrix}$ by hand.

7) Find the least squares solution to the inconsistent system $\begin{cases} x+2y=8 \\ 2x+y=12 \\ x-y=12 \end{cases}$.

8) a) Find the characteristic equation of the matrix $\begin{pmatrix} -5 & 3 \\ -9 & 7 \end{pmatrix}$.

.

b) Solve by hand for the eigenvalues and eigenvectors of this matrix.

c) Diagonalize the matrix.

9) a) Find the general solution to the differential equation $\begin{cases} \frac{dx_1}{dt} = -5x_1 + 3x_2 \\ \frac{dx_2}{dt} = -9x_1 + 7x_2 \end{cases}$

b) Solve the initial value problem if $x(0) = \begin{pmatrix} 4 \\ 6 \end{pmatrix}$

c) Carefully sketch the trajectories of the typical solution curves.

- 10) Find the general solution to the difference equation $y_{k+2} - 5y_{k+1} + 6y_k = 0$
- 11) Let $T(\mathbf{x})$ be the linear transformation defined by $\mathbf{x} \mapsto A\mathbf{x}$, where $A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \end{pmatrix}$
- a) What are the domain and codomain of T ?
- b) Determine whether T is one-to-one, onto, both, or neither.
- c) What is the kernel of T ?
- 12) Find an orthogonal basis for the space W spanned by $\left\{ \begin{pmatrix} 1 \\ 1 \\ 0 \\ 1 \\ 1 \end{pmatrix}, \begin{pmatrix} 3 \\ 5 \\ 1 \\ 3 \\ 1 \end{pmatrix}, \begin{pmatrix} 2 \\ 0 \\ 1 \\ 1 \\ 5 \end{pmatrix} \right\}$ and an orthogonal basis for $W^\perp$.
- 13) What is the distance between the point $(3, 4, 2, -4, 1)$ and the vector space W from the previous problem?
- 14) Note that if $y = Ce^{kx}$ is an exponential model for fitting data, then $\ln(y) = \ln(Ce^{kx}) = \ln(C) + kx$ gives a linear relationship between $\ln(y)$ and x . So, the least-squares line for the data $(x_0, \ln(y_0)), (x_1, \ln(y_1)), \dots$ can be used to find a best-fitting exponential curve for the (x, y) pairs. Use this to find the exponential function $y = Ce^{kx}$ which best fits the data points $(0, 2)$, $(1, 4)$, and $(2, 10)$.

15) Orthogonally diagonalize $\begin{pmatrix} 3 & 2 \\ 2 & 6 \end{pmatrix}$

16) Find the maximum value of $-3x_1^2 + 10x_1x_2 - 3x_2^2$ subject to the constraint $x_1^2 + x_2^2 = 1$

17) Prove the following is true or give a counterexample to show it may be false: The set of all 2×2 non-invertible matrices forms a subspace of $M_{2 \times 2}$.

18) Prove the following is true or give a counterexample to show it may be false: The set of all 2×2 symmetric matrices forms a subspace of $M_{2 \times 2}$